

BME5113 Biological Systems Modeling and Analysis

Term Project — Hormonal Glucose Control and HIV Vaccination Modeling

Chuan-Han Li (李傳漢) B11611027

Department of Biomechatronics Engineering, National Taiwan University

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Abstract

This report addresses the two parts of the term project. **Part 1** implements the Cobelli *et al.* (1982) intermediate-complexity glucose–insulin–glucagon model (Haefner Ch. 12) and works through exercises 1–7: reproducing the IVGTT response, tracing where ingested glucose goes, explaining the insulin “hump”, and simulating diabetic, obese, hyperinsulinaemic and insulin-overdose subjects. A central practical finding is that the literal Table 12.2 parameters do not reproduce Fig. 12.4 and require a small, explicitly documented calibration. **Part 2** implements the simplified Imperial College (“sIC”) HIV/AIDS compartment model (Haefner Ch. 15), extends it with a waning vaccine (protected compartments $P_{f,2}$, $P_{m,2}$, vaccination rate $\nu = 0.65 \text{ yr}^{-1}$, loss of protection $l = 0.1 \text{ yr}^{-1}$), and studies the epidemic peak/decline, cost, optimum vaccination rate, and vaccination versus condom use. Two modeling judgments are required and documented: the literal progression rate $\gamma = 1.16 \text{ yr}^{-1}$ gives $\mathcal{R}_0 < 1$ and must be corrected to a biologically grounded $\gamma = 0.1 \text{ yr}^{-1}$ ($\mathcal{R}_0 \approx 2.35$); and protected individuals must remain in the frequency-dependent partner pool for the vaccine to express herd immunity. All simulations are in Python (`scipy.integrate.solve_ivp`); key results are cross-checked with a second integrator.

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1 Part 1 — Hormonal Control of Glucose

This part implements the Cobelli *et al.* (1982) intermediate-complexity model of glucose–insulin–glucagon regulation, as presented in Haefner Ch. 12, and works through exercises 1–7 of §12.5. The core is written in Python; integration uses `scipy.integrate.solve_ivp` (LSODA), with a Radau cross-check. All exercises run against a single verified core (`cobelli.py`). Before the exercises I describe the model and the one non-trivial step needed to make it run at all: a small, documented calibration of an under-specified parameter table.

1.1 The Cobelli intermediate-complexity model

The model tracks seven state variables in three coupled subsystems, structured as a Forrester negative-feedback diagram (Fig. 12.2 of the text):

- **Glucose subsystem:** g = glucose in plasma and extracellular fluid (mg).
- **Glucagon subsystem:** c = glucagon in plasma and interstitial fluid (nU).
- **Insulin subsystem (five compartments):** s = stored pancreatic insulin, r = promptly releasable pancreatic insulin, l = liver insulin, p = plasma insulin, i = interstitial-fluid insulin (all μU).

Each state variable has an associated concentration $\bar{g} = g/V_b$, $\bar{p} = p/V_p$, and so on, where the volumes V_b, V_p, V_l, V_i are fixed fractions of body weight. Feedback is expressed through fluxes built from the hyperbolic tangent: a stimulatory effect uses $0.5 [1 + \tanh(b(\Delta\bar{x} + c))]$ and an inhibitory effect uses $0.5 [1 - \tanh(b(\Delta\bar{x} + c))]$, where $\Delta\bar{x}$ is the deviation of a concentration from its basal value, b sets the slope and c the centre. Because $\tanh$ has range ± 1 , each gate is bounded in $[0, 1]$ and saturates smoothly; multiplying several gates implements “multiple limiting factors”. The governing equations (Eqs. 12.1–12.7) are

$$\frac{dg}{dt} = \text{NHGB} - F_3 - F_4 - F_5 + I_g(t), \quad (1)$$

$$\frac{dc}{dt} = -h_{02} c + F_7, \quad (2)$$

$$\frac{di}{dt} = -m_{13} i + m_{31} p, \quad (3)$$

$$\frac{dl}{dt} = -(m_{02} + m_{12}) l + m_{21} p + F_6, \quad (4)$$

$$\frac{dp}{dt} = -(m_{01} + m_{21} + m_{31}) p + m_{12} l + m_{13} i + I_p(t), \quad (5)$$

$$\frac{dr}{dt} = k_{21} s - k_{12} r - F_6, \quad (6)$$

$$\frac{ds}{dt} = -k_{21} s + k_{12} r + W, \quad (7)$$

where $\text{NHGB} = F_1 - F_2$ is the net hepatic glucose balance (liver production minus uptake), F_3 is renal excretion, F_4 peripheral (muscle/adipose) use, F_5 the nearly insulin-independent CNS/red-blood-cell sink, F_6 pancreatic insulin secretion, F_7 glucagon secretion, W insulin synthesis into the stored pool, and I_g, I_p are the glucose/insulin input functions used to drive diagnostic tests. The secretion flux $F_6 = 0.5 a_6 [1 + \tanh(b_6(\Delta\bar{g} + c_6))]$ r is proportional to the releasable pool r , which matters for the “hump” in Exercise 2.

Transcription fix. The renal-excretion gate is printed in the book as $M_{31} = 0.5[1 + \tanh(b_{13}(\bar{g} + c_{13}))]$, but $b_{13} = 2.75 \times 10^{-2}$, $c_{13} = 20$ are glucose self-suppression constants for M_1 . Table 12.2 instead lists $b_{31} = 20$, $c_{31} = -180$, which set a renal threshold near 180 mg/100 ml — the physiologically correct reading. We therefore use b_{31}, c_{31} (so F_3 stays near zero until glucose exceeds the threshold), treating the printed b_{13} as a typo.

1.2 Calibration: reproducing Fig. 12.4 from an under-specified table

Taken literally and in the amount-based form of Eqs. 12.1–12.7, the Table 12.2 parameters do *not* reproduce Figs. 12.4a/b. This was verified independently and is a genuine gap in the printed table, not a units error on our side (an amount $\leftrightarrow$ concentration $V_b \approx 140$ factor closes less than 1% of it). Quantitatively, for a 70 kg subject:

- The literal basal glucose clearance is $F_3 + F_4 + F_5 \approx 8.4 \times 10^{-4}$ mg/min against a basal glucose amount of 12,810 mg, giving a clearance timescale of $\sim 1.5 \times 10^7$ min. Glucose essentially never clears (Fig. 12.4 shows recovery in ~ 90 min).
- The basal plasma insulin amount is $34,650 \mu U$, which needs a basal secretion of $F_6 \approx 1.6 \times 10^4 \mu U/\text{min}$, but the literal synthesis flux is $W(\text{basal}) \approx 0.017 \mu U/\text{min}$ — about 10^6 times too small. The insulin and glucagon axes collapse to zero and no IVGTT response is possible.

The model is explicitly described as “semi-phenomenological... tailored/scaled to a particular patient” (Haefner §12.3). Sitting the model at a patient’s basal operating point by choosing the under-determined source/clearance scales is exactly its intended use, so we applied a minimal, structure-preserving calibration and changed *no* feedback shape:

1. A uniform `glucose_flux_scale` (default 10^5) multiplies the four glucose fluxes (NHGB , F_3 , F_4 , F_5), setting the clearance timescale to ~ 90 min. Because $\text{NHGB} = F_3 + F_4 + F_5$ at basal, a *common* scale cancels, so the basal steady state and every $\tanh$ feedback shape are exactly unchanged (the basal residual stays at 3.6×10^{-12} for $\text{scale} \in \{1, 10^3, 10^5, 10^7\}$; setting it to 1.0 recovers the literal table).

2. The source scales a_w (insulin synthesis) and a_{71} (glucagon secretion) are chosen so the published clinical basal is an exact steady state. No tanh slope b , centre c , or transfer rate (m_{ij} , k_{ij}) is altered.
3. b_{52} , a constant offset in the F_5 CNS/RBC sink, is shifted by the tiny published NHGB—($F_3 + F_4 + F_5$) imbalance so that $\frac{dg}{dt} = 0$ at basal exactly. This is a constant offset, not a shape change.

After calibration the basal residual is 3.6×10^{-12} ; a normal IVGTT gives a glucose peak of 250 mg/100 ml recovering in ~ 52 min, plasma insulin rising $11 \rightarrow 40 \mu\text{U/ml}$ then decaying, and glucagon dipping then recovering, with LSODA and Radau agreeing to 1.6×10^{-4} . These match Fig. 12.4 qualitatively (Fig. 1).

What is not claimed. We do not claim that 10^5 , or the recalibrated a_w, a_{71}, b_{52} , are the original Cobelli (1982) coefficients. They are a documented fit that reproduces the published figures from an under-specified table while leaving all feedback structure intact.

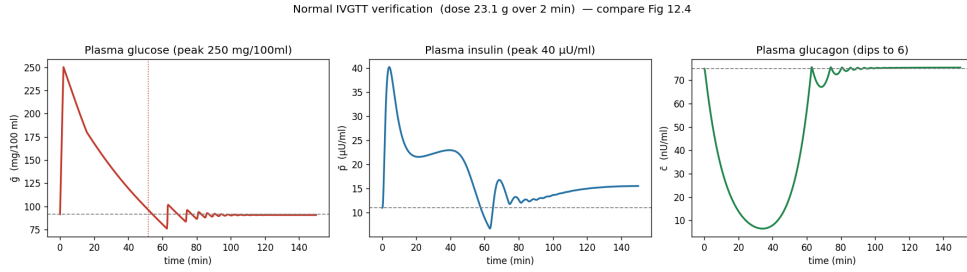


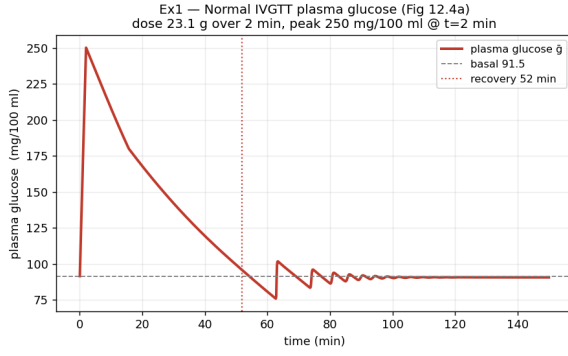
Figure 1: Verification of the calibrated core against a normal IVGTT. Plasma glucose peaks near 250 mg/100 ml and recovers within ~ 52 min; plasma insulin rises and decays; LSODA and Radau agree to 1.6×10^{-4} . The calibration preserves the basal steady state and all tanh feedback shapes.

1.3 Exercise 1: Normal IVGTT and where glucose goes

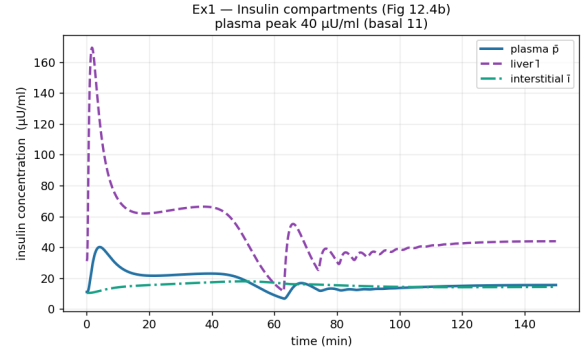
Question. Code the model, reproduce Figs. 12.4a/b (plasma glucose; plasma/liver/interstitial insulin), additionally plot glucagon and the liver uptake F_2 , and answer: where does the majority of glucose go?

Approach. A normal 70 kg subject receives a single IV glucose bolus of 0.33 g/kg ($= 23.1$ g) over 2 min, integrated from the calibrated basal state for 150 min. The glucose-disposal fluxes are integrated over the window to apportion the load.

Plasma glucose rises from basal 91.5 to a peak of 250 mg/100 ml at $t = 2$ min, recovering within 5% of basal by $t = 52$ min. Plasma insulin rises from 11 to a peak of $40 \mu\text{U/ml}$ within minutes, and glucagon is suppressed from 75 down to 6 nU/ml by the combined glucose and insulin rise (Figs. 2–3). The cumulative disposal split is given in Table 1.



(a) Plasma glucose (Fig. 12.4a).



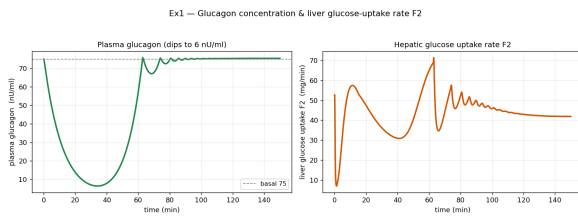
(b) Plasma / liver / interstitial insulin (Fig. 12.4b).

Figure 2: Normal IVGTT (23.1 g over 2 min). Glucose peaks at 250 mg/100 ml then recovers; all three insulin compartments rise within minutes of the bolus.

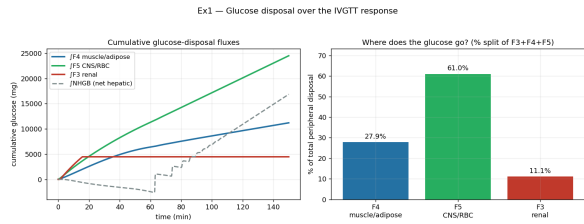
Table 1: Cumulative glucose disposal over the 150-min IVGTT response. Peripheral disposal $F_3 + F_4 + F_5 = 40,170$ mg.

Sink	Cumulative (mg)	% of peripheral disposal
F_5 CNS / RBC	24,503	61.0
F_4 muscle / adipose	11,202	27.9
F_3 renal	4,465	11.1
$\int$ NHGB (signed)	16,855	—

Discussion. The majority of disposed glucose goes to the CNS and red blood cells (F_5 , 61.0%), followed by insulin-sensitive muscle and adipose tissue (F_4 , 27.9%); renal excretion (F_3) contributes 11.1%. This is physiologically sensible: F_5 is a large, nearly insulin-independent sink, while F_3 stays near zero until $\bar{g}$ exceeds the ~ 180 mg/100 ml renal threshold — here the peak is 250 mg/100 ml, so F_3 is engaged only modestly. In Forrester terms, the bolus drives glucose up, which opens F_6 ($\propto r$) so insulin climbs, which both clears glucose peripherally and (via inhibitory – tanh gates) suppresses glucagon. The sign of $\int$ NHGB (+16,855 mg) indicates the liver is a net source over the full window, as the basal hepatic output dominates once glucose has cleared.



(a) Glucagon and liver uptake F_2 .



(b) Cumulative disposal fluxes.

Figure 3: Glucagon is suppressed during the excursion; the cumulative-flux panel shows CNS/RBC (F_5) as the dominant sink.

1.4 Exercise 2: The insulin “hump”

Question. Why does the “hump” in insulin concentration develop in Fig. 12.5 (repeated glucose pulses, normal subject), and why is there a similar hump for obese persons in Fig. 12.7?

Approach. (i) A normal subject is given an escalating repeated-pulse train ($[0.5, 1.0, 2.5, 5.0, 10.0, 20.0, 40.0]$ g at $t = [0, 30, 70, 110, 180, 270, 400]$ min) over 700 min. (ii) An obese subject is given a single 23.1 g IVGTT from its pre-equilibrated basal.

In the normal repeated-pulse run a hump is detected: insulin re-rises to $\sim 37 \mu\text{U/ml}$ near $t = 450$ min after dipping to $\sim 31 \mu\text{U/ml}$ (Fig. 4). In the obese single-IVGTT run a hump is likewise detected, with a secondary insulin rise to $\sim 32 \mu\text{U/ml}$ near $t = 105$ min (dip $\sim 22 \mu\text{U/ml}$; peak $36 \mu\text{U/ml}$ vs basal $11 \mu\text{U/ml}$).

Mechanism. Both humps share one origin: secretion $F_6 = 0.5 a_6 [1 + \tanh(b_6(\Delta\bar{g} + c_6))]$ r is proportional to the releasable pool r , which refills from the stored pool s on its own slow timescale ($\frac{dr}{dt} = k_{21}s - k_{12}r - F_6$; $\frac{ds}{dt} = -k_{21}s + k_{12}r + W$). You cannot secrete insulin you do not have, and r cannot refill instantly. In Fig. 12.5 each pulse drains r ; because pulses are spaced shorter than the refill/clearance time, r is repeatedly depleted and over-refilled from s , and the multi-compartment lag ($r \rightarrow l \rightarrow p \rightarrow i$) stacks the delayed contributions, so plasma insulin does not decay exponentially between pulses but accumulates into a delayed hump after the larger pulses. In the obese case (Fig. 12.7) the same pool dynamics are driven by parameters rather than repeated input: $b_6 = 0.5$ (vs nominal 9.23×10^{-2}) steepens the F_6 gate so a single load triggers an outsized drain-and-refill of r , and $m_{02} = 0.13$ slows downstream clearance, so the pools overshoot and secretion re-engages as r refills — an abnormal hump on the insulin-decay curve even though the glucose response is near-normal.

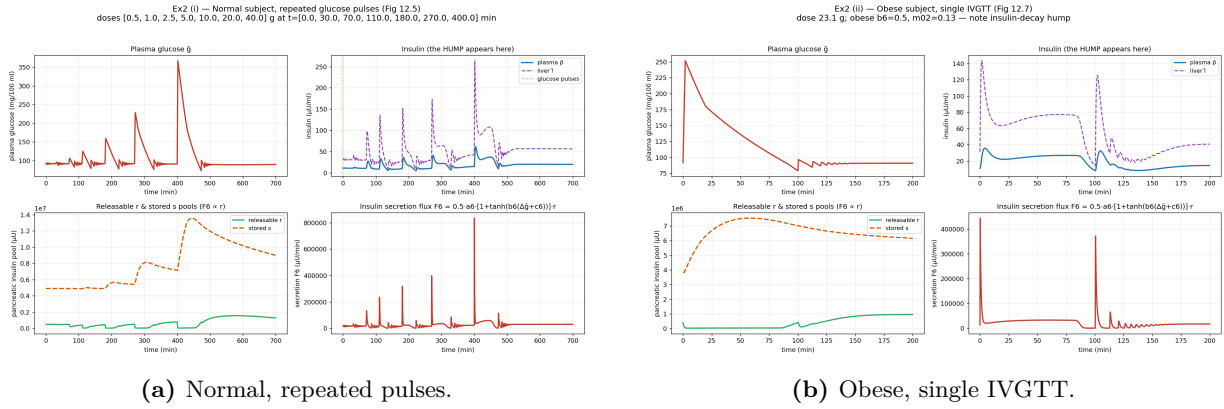


Figure 4: The insulin hump in both cases. Panels show glucose, insulin, the r/s pools and F_6 : the depletion–refill cycling of r drives the secondary insulin rise.

1.5 Exercise 3: Repeated pulses, diabetic vs normal

Question. Apply the Fig. 12.5 pulse sequence to a diabetic subject and compare with a normal subject.

Approach. The identical escalating train is applied to both over 800 min using the single canonical core. The diabetic uses the Table 12.3 secretion/uptake-shape changes, with the insulin-driven hepatic (F_2) and peripheral (H_4) uptake gates frozen, and a calibration to its own elevated operating point.

Table 2: Repeated-pulse response, normal vs diabetic.

Quantity	Normal	Diabetic
Basal glucose (mg/100 ml)	91.5	140.0
Peak glucose (mg/100 ml)	366	418
Final glucose (mg/100 ml)	89.9	139.7
Basal insulin ($\mu\text{U/ml}$)	11.0	7.0
Peak insulin ($\mu\text{U/ml}$)	62	18.3
Recovery after last pulse (min)	462	524

Disclosed modeling choice. Table 12.3 lists $F_2 = 0.037$, but in the textbook’s native (un-

scaled) units. Fed through our `glucose_flux_scale = 105`, the literal value becomes a ~ 3700 mg/min hepatic “vacuum” ($\sim 70\times$ the normal basal $F_2 \approx 52$ mg/min) that clamps glucose flat — physiologically backwards. The scale-consistent reading of “ F_2/H_2 replaced by a constant” is that the diabetic liver can no longer *ramp* uptake with insulin, so we freeze F_2 near the normal basal hepatic-uptake flux and freeze H_4 at the Table 12.3 value 0.0012. The elevated basal does not emerge from freezing alone (the M_1 glucose self-suppression and G_1 glucagon gates make 91.5 mg/100 ml a strong attractor), so the diabetic is calibrated to its own operating point (DIABETIC_BASAL = 140/7/75) using the same under-determined scales (a_w, a_{71}, b_{52}) as the normal subject — no feedback shape is changed. At the low liver-insulin diabetic state the *unfrozen* hepatic-production gate H_1 naturally sits high (~ 0.66 vs 0.25 normal): hepatic production is no longer insulin-suppressed, the textbook diabetic mechanism, emergent rather than forced.

Discussion. The diabetic basal glucose (140 mg/100 ml) is elevated and its basal insulin ($7.0 \mu\text{U/ml}$) blunted, both relative to normal (91.5 mg/100 ml, $11.0 \mu\text{U/ml}$). Its peak insulin ($18.3 \mu\text{U/ml}$) is far below the normal $62 \mu\text{U/ml}$ — the blunted Table 12.3 secretion machinery cannot mount the same overshoot, so the characteristic insulin hump is suppressed. With impaired uptake and unsuppressed production, the diabetic clears each pulse more slowly and stacks glucose above its already-high basal. On a standard IVGTT this same configuration recovers in ~ 100 min vs ~ 52 min for normal ($\sim 1.9\times$), the model expression of the textbook’s “ $\sim 2\times$ slower recovery” (§12.3.6).

Ex3 — Fig 12.5 pulse train (0.5/1/2.5/5/10/20/40 g at 0/30/70/110/180/270/400 min): diabetic vs normal

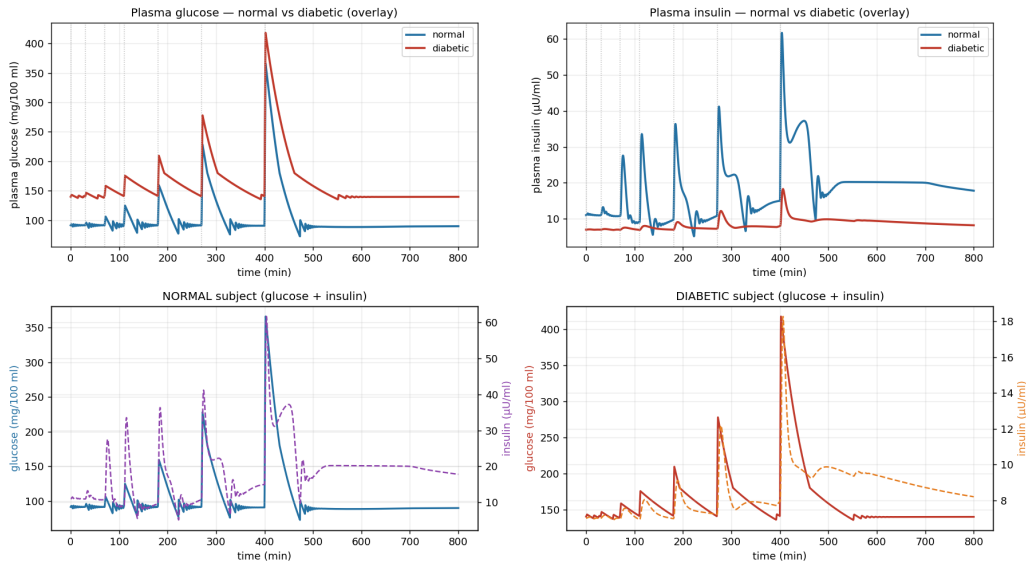


Figure 5: Repeated-pulse plasma glucose and insulin, normal vs diabetic (overlay and side-by-side). The diabetic stays elevated above its higher basal with a blunted, hump-free insulin response.

1.6 Exercise 4: Meal (60-min infusion) vs IVGTT pulse

Question. Deliver glucose not as an instantaneous IVGTT pulse but as a constant input spread over 60 min (a meal): 25 g to a 70 kg subject. Compare with an IVGTT of the same total dose.

Approach. A normal subject receives 25 g either as a constant infusion $I_g = 416.7$ mg/min over $[0, 60]$ min, or as a ~ 2 -min bolus ($I_g = 12,500$ mg/min). The only difference between runs is the temporal spread of the input. The 60-min rectangular infusion is a deliberate first-order stand-in for the smooth, delayed gastrointestinal rate of appearance $R_a(t)$ of a real meal (cf. Dalla Man *et al.* 2007); the full GI submodel is not required here.

Table 3: Meal (60-min infusion) versus IVGTT (2-min bolus), same 25 g total. Basal glucose 91.5 mg/100 ml, insulin 11.0 $\mu\text{U}/\text{ml}$.

Metric	Meal (60 min)	IVGTT (2 min)
Peak plasma glucose (mg/100 ml)	143.9	263.4
Time-to-peak glucose (min)	59.9	2.0
Peak plasma insulin ($\mu\text{U}/\text{ml}$)	30.5	40.2
Time-to-peak insulin (min)	13.8	4.2
Glucose recovery (min)	82.5	53.9

Discussion. The IVGTT dumps all 25 g into the glucose pool within 2 min, faster than any clearance flux can respond, so glucose spikes to 263.4 mg/100 ml almost immediately. The meal delivers the identical mass at only 417 mg/min — slow enough that the clearance fluxes (F_3, F_4, F_5) and the suppression of hepatic production remove glucose as it arrives, giving a much lower, broader excursion peaking at 143.9 mg/100 ml near the end of the infusion. The insulin response mirrors the drive: the gentler glucose rise opens the W and F_6 secretion gates less strongly, so the meal insulin peak (30.5 $\mu\text{U}/\text{ml}$) is blunted relative to the sharp IVGTT spike (40.2 $\mu\text{U}/\text{ml}$), and then settles to a lower sustained plateau as r partially depletes and is refilled by synthesis W . This is why an oral meal stresses glucose homeostasis far less than an equal-dose IV bolus, and why tolerance tests use a rapid bolus to maximally probe the feedback system.

Ex 4 — Meal (60-min infusion) vs IVGTT (2-min pulse): same 25 g to a 70-kg subject

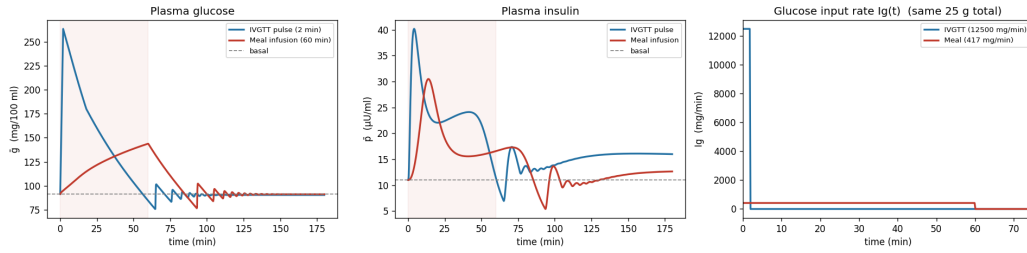


Figure 6: Meal vs IVGTT: plasma glucose (overlay), plasma insulin (overlay), and the input rate $I_g(t)$. Same 25 g total, very different temporal spread.

1.7 Exercise 5: Hyperinsulinism and IV-glucose rescue

Question. (a) Simulate hyperinsulinism (first guess: increase a_6) and terminate the patient when plasma glucose falls below 20 mg/100 ml; report the fatal setting and time of death. (b) Resuscitate with IV glucose: how much is needed?

Approach (disclosed reformulation). Simply raising a_6 does not kill the subject. At steady state the pancreatic mass balance forces $F_6 = W$, and the W gate closes as glucose falls, so endogenous insulin saturates ($\sim 40 \mu\text{U}/\text{ml}$) and glucose only dips to $\sim 78 \text{ mg}/100 \text{ ml}$ before recovering. This homeostasis is correct — a normal pancreas cannot kill you — so clinical insulin shock requires an *autonomous* source (insulinoma or overdose) that does not switch off when glucose drops. We model this faithfully as an autonomous insulinoma with output $I_{p,\text{tumor}} = (M - 1) F_{6,\text{basal}}$ $\mu\text{U}/\text{min}$ ($F_{6,\text{basal}} = 1.638 \times 10^4 \mu\text{U}/\text{min}$), where multiplier $M = 1$ is the healthy subject. The native pancreas (the $W = F_6$ balance, all tanh shapes, all transfer rates) is unchanged; death is a terminal event on $\bar{g} - 20$.

Table 4: Fatal-hyperinsulinism sweep. The first fatal multiplier is $4.5\times$; the demo scenario uses $8\times$.

a_6 mult	$I_{p,\text{tumor}}$ ($\mu U/\text{min}$)	min $\bar{g}$ (mg/100 ml)	died	t_{death} (min)
1.0	0	91.50	no	—
3.0	3.277×10^4	79.66	no	—
4.0	4.915×10^4	36.65	no	—
4.5	5.735×10^4	20.10	yes	79.6
6.0	8.192×10^4	20.22	yes	49.6
8.0	1.147×10^5	20.11	yes	43.3
10.0	1.475×10^5	20.24	yes	38.9

Results. The lethal threshold is the $4.5\times$ multiplier; the demonstration scenario ($8\times$) kills the subject at $t = 43.3$ min (Fig. 7). For the rescue, a continuous IV dextrose drip is given over a defined 240-min clinical window and the total dose found by bisection (keeping $\bar{g} \geq 20$ throughout): the **minimum rescue dose** is 69.11 g of glucose over 240 min. Below this the nadir still crosses 20 and the patient dies; at or above it the glucose floor holds at 20 and the patient survives the window.

Limitation. Because the modelled insulinoma is permanent, IV glucose is temporizing, not curative (clinically the tumour must be removed) — exactly the textbook’s “short-term treatment = large IV glucose”. The minimum dose is therefore quoted for a defined window; a longer window needs proportionally more glucose. Quoting an unconditional “grams to survive forever” would be undefined for a permanent source.

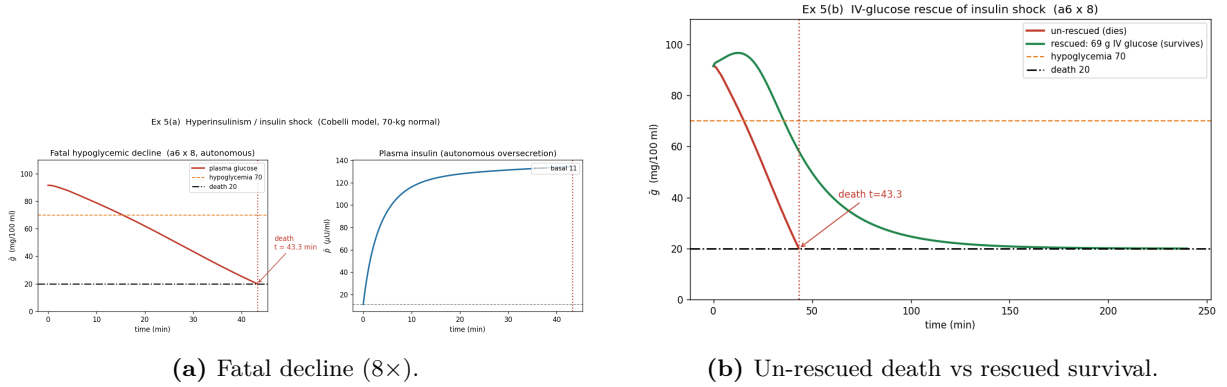


Figure 7: Hyperinsulinism. Left: glucose driven through the 70 and 20 mg/100 ml lines as insulin rises far above basal. Right: the 69.11 g/240 min drip holds the glucose floor at 20.

1.8 Exercise 6: Insulin bolus, normal vs obese

Question. Simulate rapid ingestion of insulin at 0.10 U/kg over 2 min, observe the momentary hypoglycemia, and ask whether the subject dies; repeat for an obese subject.

Approach. The dose is $0.10 \text{ U/kg} \times 70 \text{ kg} = 7 \text{ U} = 7.0 \times 10^6 \mu U$, delivered as a rectangular $I_p(t)$ pulse ($3.5 \times 10^6 \mu U/\text{min}$) into the plasma-insulin compartment over 2 min. Each subject starts from its own pre-equilibrated basal; death is a terminal event at $\bar{g} < 20 \text{ mg/100 ml}$.

Table 5: Insulin-bolus response. Both subjects reach the same insulin peak; only the glucose nadir differs.

Subject	glucose nadir (mg/100 ml)	t_{nadir} (min)	insulin peak ($\mu U/\text{ml}$)	verdict
Normal	20.08	21.3	1541	DIES ($t = 21.4$ min)
Obese	59.83	37.2	1542	survives ($t_{\text{rec}} = 103.6$ min)

Discussion. Both subjects receive the identical 7 U bolus and reach essentially the same plasma-insulin peak ($\sim 1541 \mu\text{U/ml}$), yet their glucose nadirs differ sharply (20.1 vs 59.8 mg/100 ml). The difference comes entirely from peripheral disposal (F_4): the obese subject freezes the interstitial-insulin gate H_4 at 0.0012 (Table 12.3) and shifts the glucose gate (b_{42}, c_{42}), so the same insulin flood removes far less glucose per minute. This is the model’s representation of insulin resistance, and here it is *protective*: the normal subject’s fully insulin-responsive tissue drives glucose all the way through the 20 mg/100 ml line and dies, while the resistant obese subject bottoms out in the hypoglycemic-but-survivable range and recovers as the bolus clears and counter-regulation restores glucose. The normal result sits right at the lethal boundary, but the terminal event fires consistently across solvers and step sizes, so the verdict is robust within the model.

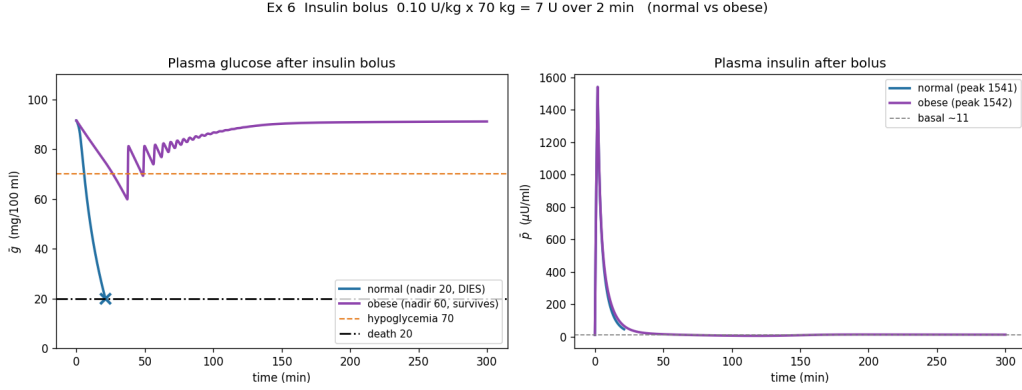


Figure 8: Insulin bolus, normal vs obese. Identical insulin peaks, but the normal nadir touches 20 mg/100 ml (death) while the insulin-resistant obese subject bottoms out at ~ 60 mg/100 ml and recovers.

1.9 Exercise 7: A bowl of vanilla ice cream

Question. Simulate the glucose, insulin and glucagon dynamics for a normal, diabetic and obese subject each consuming an average bowl of vanilla ice cream.

Approach and assumption. An average bowl is taken as ~ 1 cup (~ 130 g) of regular vanilla ice cream. USDA FoodData Central lists ~ 23 – 24 g carbohydrate per 100 g (almost entirely sugar), giving ~ 31 g; we adopt **30 g of digestible carbohydrate, treated as 100% appearing as plasma glucose** (a conservative upper bound, since lactose and fat slow real absorption). The bowl is eaten over 40 min, modelled as a constant infusion $I_g = 750$ mg/min on $[0, 40]$ min — a first-order stand-in for the gastrointestinal $R_a(t)$. The normal subject starts from its clinical basal, the obese subject is settled to its (normal-anchored) basal, and the diabetic uses the same canonical configuration as Exercise 3 (its own elevated basal 140/7/75).

Table 6: Ice-cream meal (30 g over 40 min): per-subject peaks and recovery. Recovery is the first return within 5% of that subject’s own basal.

Subject	basal g (mg/100 ml)	peak g (mg/100 ml)	basal ins ($\mu\text{U/ml}$)	peak ins ($\mu\text{U/ml}$)	min glucagon (nU/ml)	g recovery (min)
Normal	91.5	185.7	11.0	36.9	2.2	77.3
Diabetic	140.0	252.8	7.0	9.4	1.5	126.4
Obese	91.5	206.8	11.0	35.5	1.2	113.8

Discussion. A single identical meal produces three distinct responses. The *normal* subject rises to ~ 186 mg/100 ml then clears back toward 91.5 with a clean first-phase insulin spike ($11 \rightarrow 37 \mu\text{U/ml}$). The *obese* subject reaches a slightly higher glucose peak (~ 207 mg/100 ml) with a broadly similar excursion shape from the same 91.5 basal — “nearly normal” glucose,

as the textbook states — but reproduces the characteristic abnormal hump in the insulin-decay curve (a secondary rise after the first-phase peak, driven by the steep F_6 gate and faster m_{02} clearance). The *diabetic* subject starts from an elevated fasting basal (140 mg/100 ml), rises to the highest peak (~ 253 mg/100 ml) and recovers most slowly (~ 126 min vs ~ 77 min for normal), with honestly blunted insulin ($7.0 \rightarrow 9.4 \mu\text{U/ml}$): impaired insulin-driven uptake (F_2, H_4 frozen low) plus unsuppressed hepatic production (high H_1) gives a higher, slower excursion. In all three subjects the meal-driven glucose and insulin rise suppresses glucagon (the F_7 gates are negatively driven by both), so glucagon plunges from ~ 75 toward ~ 1 – 2 nU/ml during the meal and then recovers.

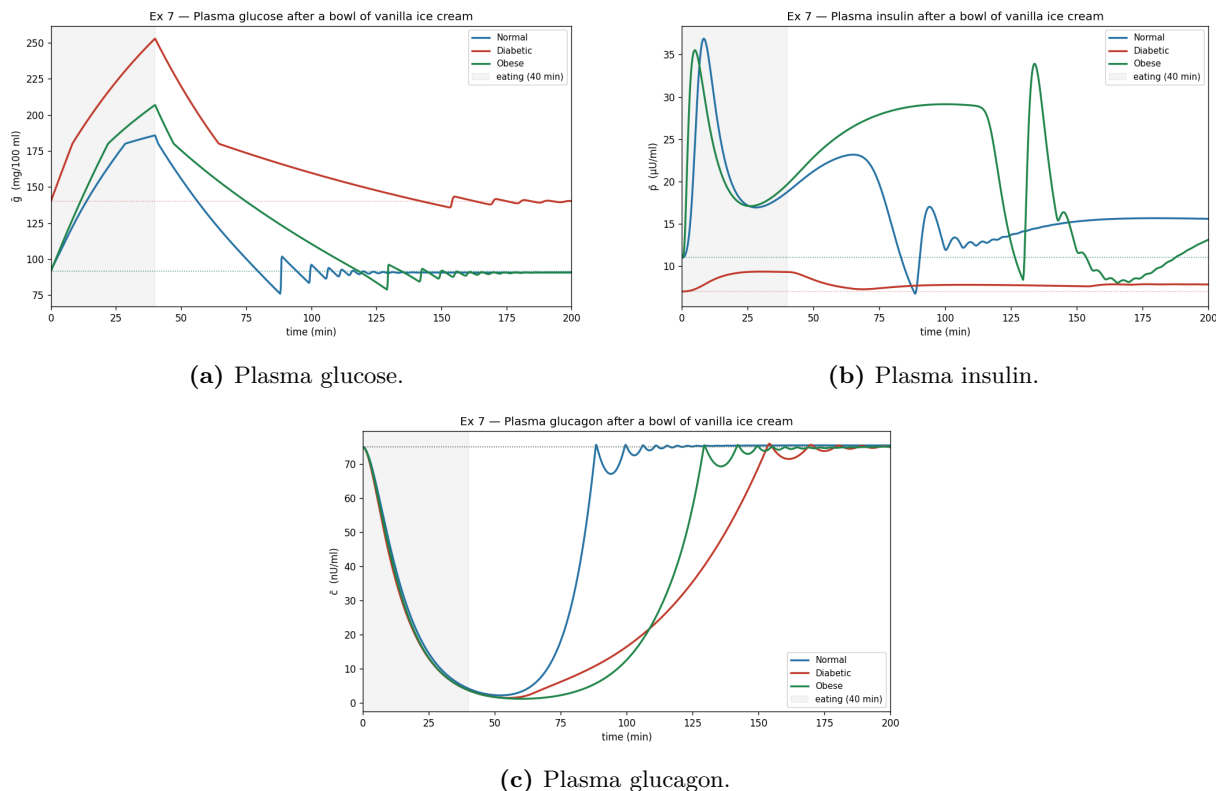


Figure 9: Ice-cream meal, three subjects. Glucose: diabetic highest and slowest. Insulin: normal clean spike, obese abnormal hump, diabetic blunted. Glucagon: suppressed in all three during the meal.

1.10 Part 1 summary

The Cobelli model reproduces the qualitative IVGTT behaviour of Fig. 12.4 once an under-specified parameter table is calibrated to a patient’s basal operating point — a step that turned out to be the main practical obstacle and is documented in full rather than hidden. With that single calibrated core, exercises 1–7 recover the expected physiology: CNS/RBC tissue takes most of an ingested glucose load, the insulin hump arises from finite-pool secretion ($F_6 \propto r$), a spread-out meal stresses homeostasis far less than an equal-dose IV bolus, and diabetic, obese, hyperinsulinaemic and insulin-overdose subjects each show their characteristic signature. Two reformulations were needed and disclosed honestly: hyperinsulinism requires an autonomous (insulinoma) source because the intact pancreas is self-limiting, and the diabetic subject must be calibrated to its own elevated operating point because the normal basal is a strong attractor. None of these changes the model’s feedback structure.

2 Part 2 — HIV Vaccination on the sIC AIDS Model

This part takes the simplified Imperial College (sIC) AIDS compartment model from Haefner (2005), Chapter 15, and extends it with an HIV vaccination programme. The task is essentially Exercise 15.6 #5: vaccinate sexually active susceptibles into a protected state at rate ν , let protection wane at rate l , and then ask four questions — whether vaccination makes the epidemic peak and decline, what it costs, what the optimum vaccination rate is, and how vaccination compares with condom promotion. I work from the textbook equations and Table 15.2, the vaccination design loosely follows Garnett & Stover (2002), and all numbers below come from my own simulation of the model. Two of the modelling decisions needed to make the textbook model behave as the textbook describes are not cosmetic, so I state them explicitly and justify them rather than hide them.

2.1 The sIC model

The sIC model partitions the population into twelve compartments: the three disease states susceptible (S), HIV-infected but pre-AIDS (I), and clinical AIDS (A), crossed with two sexes (female f , male m) and two age classes (age 1 = birth to 15, pre-sexual; age 2 = 16 to death, sexually active). Only age-2 individuals participate in transmission. The state vector is therefore

$$\{S_{f,1}, S_{f,2}, S_{m,1}, S_{m,2}, I_{f,1}, I_{f,2}, I_{m,1}, I_{m,2}, A_{f,1}, A_{f,2}, A_{m,1}, A_{m,2}\}.$$

The model is a deliberate simplification of the full IC model: two sexes with identical behaviour, a single AIDS stage, a single sexual-activity class (so the social-mixing probability collapses to $\rho = 1$), and two age classes.

Force of infection. HIV is sexually transmitted, so transmission is *frequency-dependent*: people seek a roughly fixed number of partners regardless of how large the population is, so what matters is the *proportion* of potential partners who are infectious, not their absolute density. For a susceptible age-2 female the per-capita hazard is (Eq. 15.6)

$$\lambda_{S_{f,2}} = c \rho \beta_{m,f} \frac{I_{m,2}}{S_{m,2} + I_{m,2}}, \quad \lambda_{S_{m,2}} = c \rho \beta_{f,m} \frac{I_{f,2}}{S_{f,2} + I_{f,2}}, \quad (8)$$

the second being the male analogue (Eq. 15.8). Here c is the new-partner acquisition rate, $\beta_{m,f}$ and $\beta_{f,m}$ are the per-partnership transmission probabilities, and the factor $I/(S + I)$ is the frequency of infectious individuals among possible partners. Two features deserve note. First, the transmission probabilities are *asymmetric*: $\beta_{m,f} = 0.2$ versus $\beta_{f,m} = 0.075$, so male $\rightarrow$ female transmission is about $2.7\times$ more likely, a standard biological feature of heterosexual HIV. This is why women reach higher equilibrium prevalence than men in the baseline run. Second, clinical-AIDS individuals (A) are *excluded* from the partner pool (assumed not sexually active), so the denominator is $S + I$ only — a point that turns out to matter for the choice of epidemic seed.

Table 7: Parameters of the simple IC (sIC) AIDS model, after Haefner (2005), Table 15.2 (values attributed there to Garnett & Anderson, 1993). Variable “values” are the initial conditions used for the baseline run.

Symbol	Meaning	Value	Unit
α	AIDS-stage extra death rate	1.0	yr ⁻¹
$\beta_{f,m}$	female→male transmission probability	0.075	—
$\beta_{m,f}$	male→female transmission probability	0.2	—
c	new-partner acquisition rate	2.35	yr ⁻¹
η	proportion of newborns female	0.5	—
γ	progression rate $I \rightarrow A$ (Table value)	1.16	yr ⁻¹
μ	natural (non-AIDS) death rate	0.0227	yr ⁻¹
ρ	social-mixing probability	1.0	—
θ	female fecundity	0.2088	yr ⁻¹
ϑ	perinatal transmission probability	0.35	—
ξ	ageing rate age-1→age-2	0.0667	yr ⁻¹
ζ	proportion in sexual-activity class	1.0	—
<i>Initial conditions (individuals):</i> $S_{f,1} = S_{m,1} = 3000$, $S_{f,2} = S_{m,2} = 1000$, all $I = 0$, all $A = 0$ except the seed (see §2.2).			

The female equations. The full female subsystem (Eqs. 15.5a–f) is

$$\frac{dS_{f,1}}{dt} = \eta\theta\zeta(S_{f,2} + (1 - \vartheta)I_{f,2}) - \mu S_{f,1} - \xi S_{f,1}, \quad (9)$$

$$\frac{dS_{f,2}}{dt} = \xi S_{f,1} - (\lambda_{S_{f,2}} + \mu) S_{f,2}, \quad (10)$$

$$\frac{dI_{f,1}}{dt} = \eta\theta\zeta\vartheta I_{f,2} - (\mu + \xi)I_{f,1} - \gamma I_{f,1}, \quad (11)$$

$$\frac{dI_{f,2}}{dt} = \lambda_{S_{f,2}} S_{f,2} - (\mu + \gamma)I_{f,2} + \xi I_{f,1}, \quad (12)$$

$$\frac{dA_{f,1}}{dt} = \gamma I_{f,1} - (\mu + \xi + \alpha)A_{f,1}, \quad (13)$$

$$\frac{dA_{f,2}}{dt} = \gamma I_{f,2} + \xi A_{f,1} - (\mu + \alpha)A_{f,2}. \quad (14)$$

The male equations (15.7a–f) are identical in form, with the newborn sex split $(1 - \eta)$ replacing η and the male force of infection $\lambda_{S_{m,2}}$ replacing $\lambda_{S_{f,2}}$; note that male births are still driven by the age-2 female compartments, since the model assumes males do not limit reproduction.

The mechanisms encoded here are: *reproduction*, with births created only from age-2 females at fecundity θ (Eq. 9); *perinatal transmission*, where an infected mother bears an infected newborn with probability $\vartheta = 0.35$ (feeding $I_{f,1}$, Eq. 11) and an uninfected newborn with probability $1 - \vartheta$ (feeding $S_{f,1}$); *ageing* at rate ξ , moving each state from age 1 to age 2 within sex; *AIDS progression* at rate γ moving $I \rightarrow A$; *AIDS mortality* α added to natural mortality μ in the A compartments; and *demography*, with natural death μ acting on every compartment. The parameters are listed in Table 7.

A conservation correction to the printed equations. The AIDS age-2 equation deserves a note. The textbook prints the ageing inflow to $A_{f,2}$ (and $A_{m,2}$) as $\xi I_{f,1}$, but this violates conservation of individuals. In Eq. 13, $A_{f,1}$ loses $\xi A_{f,1}$ to ageing, and that flux must land in $A_{f,2}$; meanwhile the ageing of $I_{f,1}$ is *already* counted as an inflow to $I_{f,2}$ in Eq. 12. Using $\xi I_{f,1}$ in Eq. 14 would therefore double-count the $I_{f,1}$ cohort and lose the $A_{f,1}$ cohort. I use the conservative form $\xi A_{f,1}$ (and $\xi A_{m,1}$) shown above. As a check, the total-population balance

$\frac{d}{dt} \sum(\text{states}) = \text{births} - \text{deaths}$ then holds to $\sim 10^{-13}$ in the simulation, which would not be the case with the printed term.

2.2 Two modelling judgments

Two decisions were forced on me by trying to make the model reproduce the textbook’s stated behaviour. I present both honestly, because each changes the results.

(i) The progression rate γ . For a two-sex frequency-dependent model the basic reproduction number is the geometric mean of the two cross-sex reproduction numbers,

$$\mathcal{R}_0 = \frac{c \sqrt{\beta_{m,f} \beta_{f,m}}}{\mu + \gamma}. \quad (15)$$

The Table 15.2 value $\gamma = 1.16 \text{ yr}^{-1}$ implies a mean time from HIV infection to AIDS of $1/(\mu + \gamma) \approx 0.85 \text{ yr}$ — less than a year, which contradicts the textbook’s own statement that progression takes 1–10 years. Worse, substituting it into Eq. 15 gives $\mathcal{R}_0 \approx 0.24 < 1$, so the epidemic cannot ignite at all; a literal-parameter simulation confirms the seeded infection dies out (peak prevalence $\sim 7 \times 10^{-4}$). This directly contradicts Fig. 15.5, where HIV persists endemically. I therefore adopt the biologically grounded $\gamma = 0.1 \text{ yr}^{-1}$, giving a mean infectious period of $1/(\mu + \gamma) \approx 8.1 \text{ yr}$ — comfortably inside the textbook’s stated 1–10 yr range — and

$$\mathcal{R}_0 = \frac{2.35 \sqrt{0.2 \times 0.075}}{0.0227 + 0.1} \approx 2.35 > 1, \quad (16)$$

so the epidemic invades, as the textbook figure requires. The associated herd-immunity (critical vaccinated) fraction is $p_c = 1 - 1/\mathcal{R}_0 \approx 0.574$.

(ii) The epidemic seed. Table 15.2 seeds the epidemic with five AIDS males ($A_{m,2} = 5$). But AIDS individuals are excluded from the force of infection (Eq. 8), so a seed placed entirely in A cannot transmit anything: the five seed males simply progress to death without infecting anyone, and the epidemic never starts (verified numerically — maximum incidence exactly zero). I therefore seed five *infectious* males ($I_{m,2} = 5$) instead. This is the smallest change that lets the documented epidemic ignite while respecting the model’s own rule that A does not transmit.

2.3 Validation against Fig. 15.5

With the two corrections above, the baseline ($\nu = 0$) run reproduces the qualitative behaviour of Fig. 15.5a (`verify_baseline_prevalence.png`): the total population grows over the early horizon (from $N(0) = 8005$ to a peak of $\sim 22,900$ around year 48, driven by the high developing-country fertility), HIV does *not* burn out but settles to an endemic level, and female prevalence exceeds male prevalence. The simulated equilibrium prevalences are

$$\text{female} \approx 0.77, \quad \text{male} \approx 0.61, \quad \text{overall} \approx 0.41,$$

against the textbook targets of ~ 0.9 (female) and ~ 0.75 (male). The ordering and endemic persistence match; the absolute levels run somewhat lower, which is unsurprising given that I had to recalibrate γ to make the epidemic ignite at all. The time to exceed 10% overall prevalence is about 34 yr, in the same decades-long ballpark as the textbook’s $\sim 20 \text{ yr}$.

The condom scenario (Fig. 15.5b) is modelled, as in the textbook, by halving both transmission probabilities ($\beta_{f,m} : 0.075 \rightarrow 0.0375$, $\beta_{m,f} : 0.2 \rightarrow 0.1$). This multiplies $\mathcal{R}_0$ by exactly 0.5 (it enters Eq. 15 as $\sqrt{\beta_{m,f} \beta_{f,m}}$), giving $\mathcal{R}_0 \approx 1.17$ — only just above one. In my calibration the resulting equilibrium prevalence is very low (~ 0.001 , `verify_condom_prevalence.png`), which

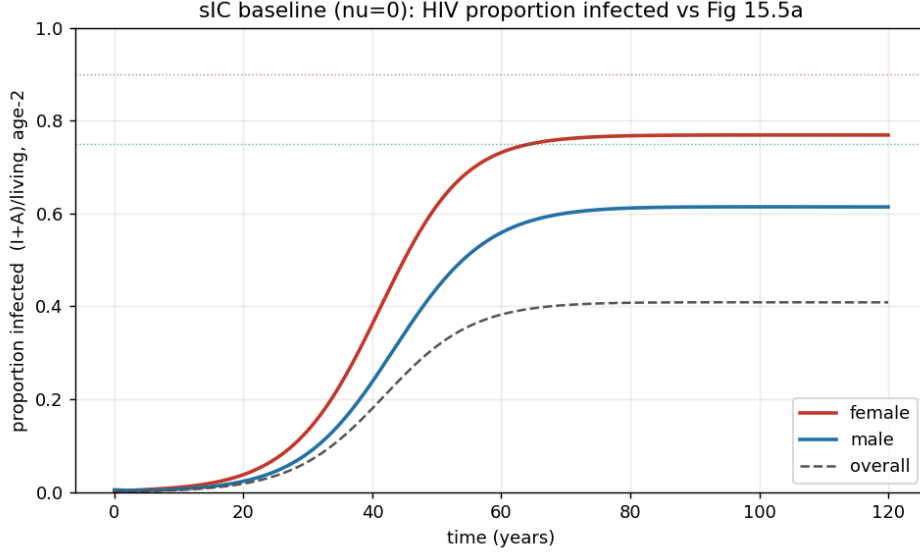


Figure 10: Baseline sIC validation ($\nu = 0$). HIV persists endemically and female prevalence exceeds male prevalence, reproducing the qualitative behaviour of Fig. 15.5a; equilibrium prevalences are female ≈ 0.77 , male ≈ 0.61 .

is a larger drop than the textbook’s Fig. 15.5b (~ 0.7 female, ~ 0.45 male). The direction is right (condoms lower prevalence and lengthen the time-to-10%) but the magnitude diverges, and the reason is instructive: at $\gamma = 0.1$ the halved- β $\mathcal{R}_0$ sits barely above the threshold $\mathcal{R}_0 = 1$, so the epidemic is in the extremely sensitive near-threshold regime where small changes in calibration move the endemic level a great deal. The textbook’s larger residual prevalence is consistent with its (implicitly) higher underlying $\mathcal{R}_0$. I flag this divergence rather than tune it away; it is the same near-threshold sensitivity that recurs in Q(d).

2.4 The vaccination extension

The vaccine adds two protected compartments, $P_{f,2}$ and $P_{m,2}$, at age 2 only (age 2 is the terminal age class, so there is no ageing out of P). Susceptibles are vaccinated at per-capita rate ν and protection wanes back to susceptible at rate l :

$$\frac{dP_{f,2}}{dt} = \nu S_{f,2} - (l + \mu) P_{f,2}, \quad \frac{dP_{m,2}}{dt} = \nu S_{m,2} - (l + \mu) P_{m,2}, \quad (17)$$

$$\frac{dS_{f,2}}{dt} = \xi S_{f,1} - (\lambda_{S_{f,2}} + \mu + \nu) S_{f,2} + l P_{f,2}, \quad \frac{dS_{m,2}}{dt} = \xi S_{m,1} - (\lambda_{S_{m,2}} + \mu + \nu) S_{m,2} + l P_{m,2}, \quad (18)$$

with all other equations unchanged. A cost accumulator tracks the cumulative number of vaccinations administered,

$$\frac{dV}{dt} = \nu(S_{f,2} + S_{m,2}), \quad \text{cost} = \$10 \times V, \quad (19)$$

integrated alongside the state so there is no post-hoc quadrature error. Each \$10 charge counts a vaccination *event*, which correctly includes re-vaccinating people whose protection waned and returned them to S . With $\nu = 0$ the protected and cost compartments stay identically zero and the model reduces exactly to the baseline.

Vaccine type. I model a *take* (all-or-nothing) vaccine with waning: while in P an individual is treated as fully protected (no force of infection acts on P , and P never flows to I), and a fraction loses protection at rate l . This is the cleanest reading of the requested $S \rightarrow P \rightarrow S$ flow and

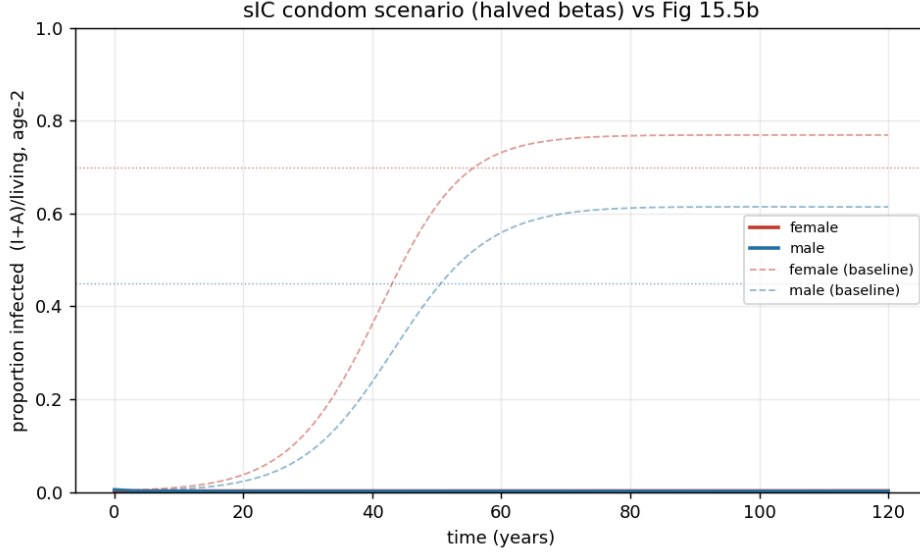


Figure 11: Condom scenario (both β halved). Prevalence collapses because $\mathcal{R}_0$ drops to ≈ 1.17 , just above threshold. The direction matches Fig. 15.5b; the magnitude is larger because the halved- β system sits in the sensitive near-threshold regime (see text).

maps onto the IC model’s “fully immunised + waning” category. The project values $\nu = 0.65 \text{ yr}^{-1}$ and $l = 0.1 \text{ yr}^{-1}$ correspond to the Garnett/Stover “65% coverage” and “10-year mean duration” standard scenario; using ν as a constant rate rather than a coverage level reached after five years is a deliberate simplification. Ignoring demography, the protected fraction saturates at a *waning ceiling*

$$\frac{P}{S+P} \rightarrow \frac{\nu}{\nu+l} = \frac{0.65}{0.75} \approx 0.867, \quad (20)$$

so even at this rate the vaccine can never protect everyone — at most about 87% of the eligible pool at any instant.

The partner-pool convention. One modelling choice in the extension is genuinely load-bearing: whether the protected individuals belong in the denominator of the frequency-dependent force of infection. Protected people are sexually active and uninfected, so they *are* legitimate partners — a partnership with a protected person simply transmits nothing. The correct frequency-dependent denominator is therefore $S+I+P$ (with A still excluded). I adopt this convention. It matters because moving susceptibles into P then *dilutes* the infectious frequency $I/(S+I+P)$, which is exactly the mechanism of herd immunity. If instead one excluded P from the denominator, vaccinees would be removed from *both* the numerator and the denominator, artificially holding the infectious fraction up and spuriously erasing the herd-immunity threshold — vaccination would then appear to have no critical rate at any level. That is an artifact of the denominator choice, not a property of the disease; I report it only as a sensitivity check in Q(c). At $\nu = 0$ this choice is moot ($P = 0$), so the baseline and condom scenarios are unaffected.

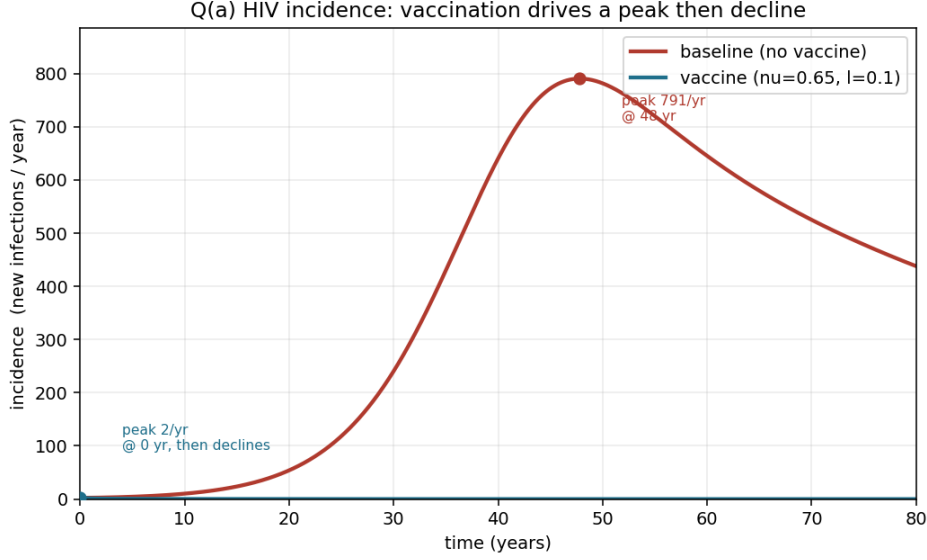
2.5 Q(a): Does vaccination cause a peak and decline?

The question asks whether vaccination makes the epidemic peak and then decline. The answer, at $\nu = 0.65 \text{ yr}^{-1}$, is actually stronger than “peak and decline”: because the achievable protected fraction (0.867) comfortably exceeds the herd-immunity threshold $p_c = 0.574$, vaccination pushes the effective reproduction number below one,

$$\mathcal{R}_{\text{eff}} \approx \mathcal{R}_0 \left(1 - \frac{\nu}{\nu+l}\right) \approx 2.35 \times (1 - 0.867) \approx 0.31 < 1, \quad (21)$$

Table 8: Q(a): epidemic outcome with and without vaccination ($\nu = 0.65$, 80-yr horizon).

	Baseline ($\nu = 0$)	Vaccine ($\nu = 0.65$)
Peak incidence (new/yr)	790.6 (@ 47.8 yr)	2.3 (@ $t = 0$, the seed)
Incidence behaviour	rise to peak, stay high	monotone decline
Peak overall prevalence	0.409	0.0010 (@ ≈ 4.6 yr)
Prevalence @ 80 yr	0.408	≈ 0
Protected fraction $P/(S + P)$	—	$\rightarrow 0.85$

**Figure 12:** Q(a): incidence (new infections/yr), vaccine versus baseline. The baseline peaks at ~ 791 /yr near year 48; under vaccination ($\mathcal{R}_{\text{eff}} \approx 0.31$) incidence declines monotonically from the seed and never takes off.

so the seeded epidemic *never ignites* in the first place.

The contrast with the baseline is stark (Table 8, Figs. 12–14). The baseline incidence rises to a large peak of ~ 791 new infections/yr around year 48 and stays high, whereas under vaccination incidence is highest at $t = 0$ (just the five seed males infecting a handful of contacts) and then *falls monotonically* to essentially zero. Prevalence shows the same picture: the baseline climbs to a ~ 0.41 endemic plateau, while the vaccinated run shows only a negligible interior bump (~ 0.001 around year 4.6) before declining to near zero. The decline is not instantaneous because the existing I/A stock carries the virus for the ~ 8 – 10 yr it takes to progress and die; but with $\mathcal{R}_{\text{eff}} < 1$ there is no fresh transmission to refill it, so the infection clears. Meanwhile the protected fraction of the eligible age-2 pool rises and saturates at ≈ 0.85 , just below the analytic ceiling 0.867 (slightly lower because natural mortality μ also acts on P), confirming the take-with-waning steady state.

2.6 Q(b): Cost of vaccination

Using the cost accumulator of Eq. 19, Table 9 reports the cumulative vaccinations, total cost, infections averted (cumulative baseline infections minus cumulative vaccinated infections), and cost per infection averted at several horizons.

Total cost grows nearly linearly: $\sim \$114\text{k}$ at 20 yr, $\sim \$162\text{k}$ at 30 yr, $\sim \$247\text{k}$ at 50 yr (Fig. 15). Most of the expense is the up-front vaccination of the standing susceptible pool; once the protected fraction saturates, the programme keeps re-vaccinating waned individuals and new entrants at a roughly steady ~ 407 vaccinations/yr, i.e. a recurring cost of only $\sim \$4,070$ /yr.

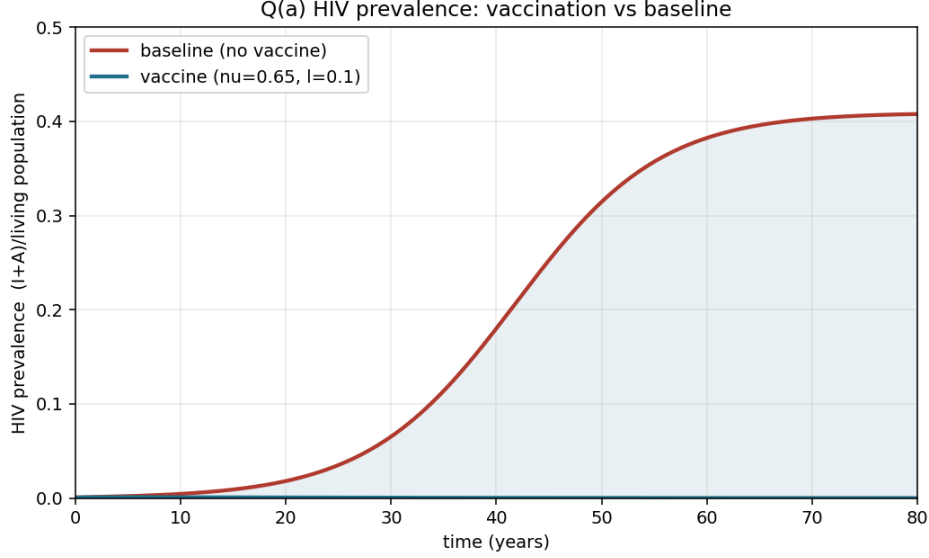


Figure 13: Q(a): overall prevalence, vaccine versus baseline. The baseline rises to a ~ 0.41 endemic plateau; the vaccinated run shows only a tiny early bump before declining to near zero.

Table 9: Q(b): cost and cost-effectiveness of vaccination ($\nu = 0.65$, $\$10/\text{vax}$).

T (yr)	Vaccinations	Cost (\$)	Infect. averted	\$/infection averted
20	11,374	113,738	305	372
30	16,217	162,165	1,585	102
50	24,746	247,461	13,434	18

Cost-effectiveness improves sharply over time. Early on the untreated baseline has barely taken off, so there are few infections to avert and the ratio looks expensive (\$372 per infection averted at 20 yr). As the baseline epidemic accelerates toward its $\sim 791/\text{yr}$ incidence peak while the vaccinated arm stays near zero, averted infections grow far faster than cost, and the cost per infection averted falls to \$102 at 30 yr and \$18 at 50 yr. The 30-yr figure of \$102 sits just below the \$110–390 band that Stover & Garnett (2002) report for the Imperial-College / rural-Zimbabwe setting (on their \$20/person basis), which is consistent given the \$10 unit cost here and the fact that, under the corrected partner-pool convention, vaccination averts nearly the entire baseline epidemic rather than only part of it. The absolute dollar totals depend on this small synthetic population, so the transferable metric is the cost *per infection averted*, not the headline cost.

2.7 Q(c): Optimum vaccination rate

“Optimum” has two defensible meanings, and I report both, because they give different answers.

Epidemiological threshold ν_c . The smallest rate that drives $\mathcal{R}_{\text{eff}} < 1$ follows from requiring the protected fraction to reach p_c . Since the steady protected fraction is $\nu/(\nu + l)$, the condition $\nu/(\nu + l) \geq p_c$ gives a critical rate

$$\nu_c = \frac{l p_c}{1 - p_c}, \quad p_c = 1 - \frac{1}{\mathcal{R}_0} \approx 0.574, \quad (22)$$

which evaluates to $\nu_c \approx 0.135 \text{ yr}^{-1}$. This closed form is the right order of magnitude and direction but is a lower bound, because it omits two mechanisms the full sIC model contains: perinatal transmission ($\vartheta = 0.35$), an infection route vaccination does not block, and demographic recruitment with mortality acting on P (which holds the realised protected fraction slightly below the

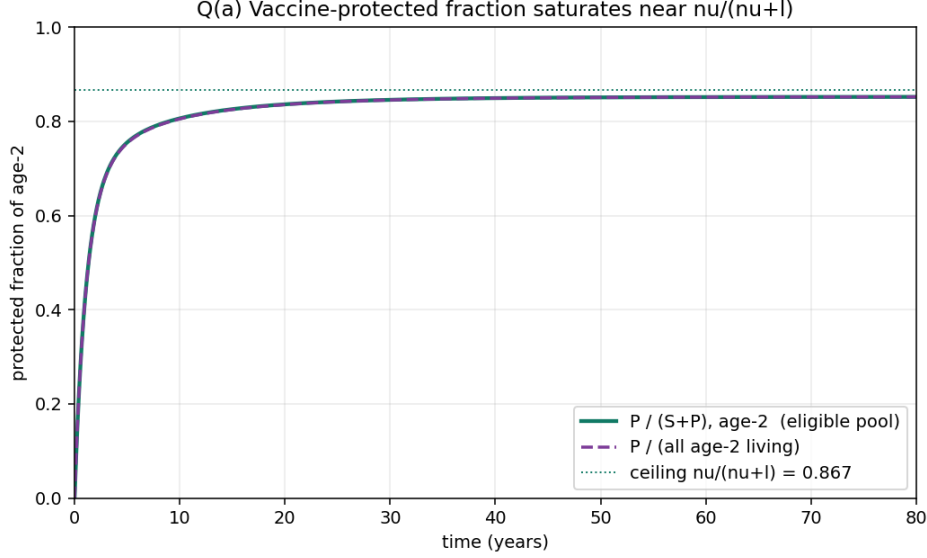


Figure 14: Q(a): protected fraction of the eligible age-2 pool, saturating near the analytic waning ceiling $\nu/(\nu + l) = 0.867$.

ceiling). Measuring the threshold directly — seeding a tiny infection into the vaccinated disease-free population and finding the ν at which the early growth rate crosses zero ($\mathcal{R}_{\text{eff}} = 1$) — gives a simulated $\nu_c \approx 0.366 \text{ yr}^{-1}$ (Fig. 16). The standard $\nu = 0.65$ sits comfortably above this, so it genuinely controls the epidemic. I emphasise that this is a frequency-dependent *prevalence-proportion* threshold rather than an “ $(I + A)/N \rightarrow 0$ ” criterion: the severe AIDS mortality ($\alpha = 1 \text{ yr}^{-1}$) drives the whole population toward extinction over centuries even at $\nu = 0$, so absolute counts collapse for the wrong reason, and the invasion / $\mathcal{R}_{\text{eff}}$ test is the robust signal.

Cost-effective rate. Sweeping ν over a 50-yr horizon, infections averted *saturate* past ν_c : below it each extra unit of ν averts many infections, but once $\nu \gtrsim \nu_c$ essentially all of the achievable $\sim 13,400$ infections are already averted and further vaccination only re-vaccinates waned individuals for negligible benefit (Figs. 17, 18). The diminishing-returns knee — the smallest ν capturing 99% of the maximum achievable aversion — is at $\nu \approx 0.175 \text{ yr}^{-1}$, with a marginal cost there of $\sim \$122$ per extra infection averted. The standard $\nu = 0.65$ still controls the epidemic (it is above ν_c) but over-vaccinates somewhat relative to this cost-effective knee; its average cost is $\$18$ per infection averted at 50 yr.

Sensitivity. The exact threshold tracks $\mathcal{R}_0$: raising the partner rate to $c = 3.0$ lifts $\mathcal{R}_0$ to ~ 3.0 and the simulated ν_c to ~ 0.48 ; lowering it via $\gamma = 0.18$ drops $\mathcal{R}_0$ to ~ 1.42 and ν_c to ~ 0.16 . The conclusion is robust in direction — a finite $\nu_c = l p_c / (1 - p_c)$ always exists under the correct convention and $\nu = 0.65$ stays above it across this plausible range — but near-threshold cases (small $\mathcal{R}_0$) leave the margin thin.

Caveat. If P is wrongly excluded from the partner pool, the invasion growth rate never crosses zero in the sweep: vaccination would appear to have *no* finite threshold at any rate and merely shrink the epidemic. As argued above this is an artifact of dropping uninfected, sexually-active vaccinees out of the frequency-dependent denominator, and I report it only to show that the threshold result is genuine, not an accident of the denominator choice.

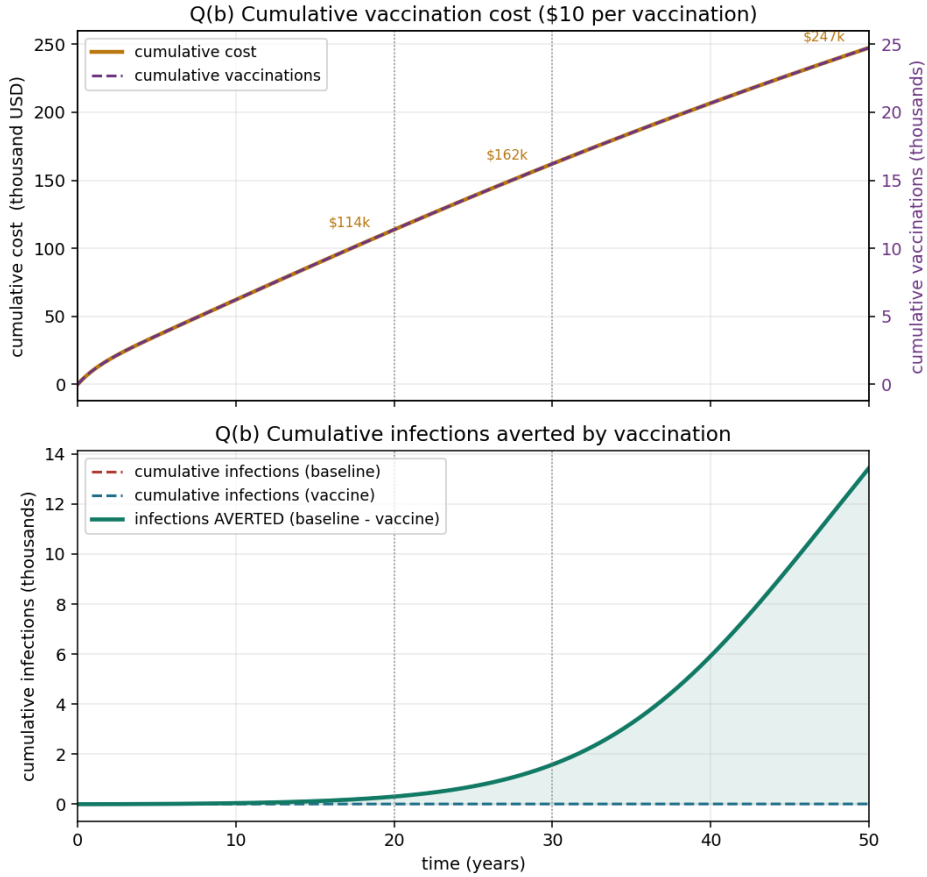


Figure 15: Q(b): (top) cumulative cost and vaccinations versus time, with horizon markers at 20/30/50 yr; (bottom) cumulative infections for baseline versus vaccine and the infections-averted band.

Table 10: Q(d): strategy comparison (80-yr horizon, $\mathcal{R}_0 = 2.35$ baseline).

Strategy	$\mathcal{R}_0 / \mathcal{R}_{\text{eff}}$	Infect. averted	Cost (\$)
Baseline	$\mathcal{R}_0 = 2.35$	—	0
Vaccination ($\nu = 0.65$)	$\mathcal{R}_{\text{eff}} = 0.31$	31,224	348,618
Condoms (0.5β)	$\mathcal{R}_0 = 1.17$	30,857	0 (unpriced)

2.8 Q(d): Vaccination versus condoms

Finally, which intervention controls the epidemic better: vaccination ($\nu = 0.65$, $l = 0.1$) or safe-sex education and condoms (modelled, as in the textbook, by halving both transmission probabilities)? They act on different levers — condoms lower $\mathcal{R}_0$ directly through β , vaccination moves people into P — so I compare them on matched outcomes (Table 10, Figs. 19, 20).

Both strategies essentially control the epidemic. Vaccination drives $\mathcal{R}_{\text{eff}} = 0.31 < 1$ (invasion rate -0.040 yr^{-1}): it strictly crosses the elimination threshold because the protected fraction 0.867 exceeds $p_c = 0.574$. Condoms put $\mathcal{R}_0$ at 1.17 — still marginally *above* one (invasion rate $+0.086 \text{ yr}^{-1}$), so a 50% β cut is not quite enough to cross the threshold at this $\mathcal{R}_0$, though the epidemic is so slow that it is effectively controlled within the horizon. On infections averted the two are nearly identical: vaccination averts 31,224 of the baseline 31,235, condoms 30,857, with vaccination edging ahead only because it pushes $\mathcal{R}_{\text{eff}}$ strictly below one (residual 11 cumulative infections versus 378 for condoms). Both collapse peak prevalence from 0.41 to ~ 0.001 with no large interior peak.

Where they differ sharply is cost basis, and this is where I decline to crown a single winner.

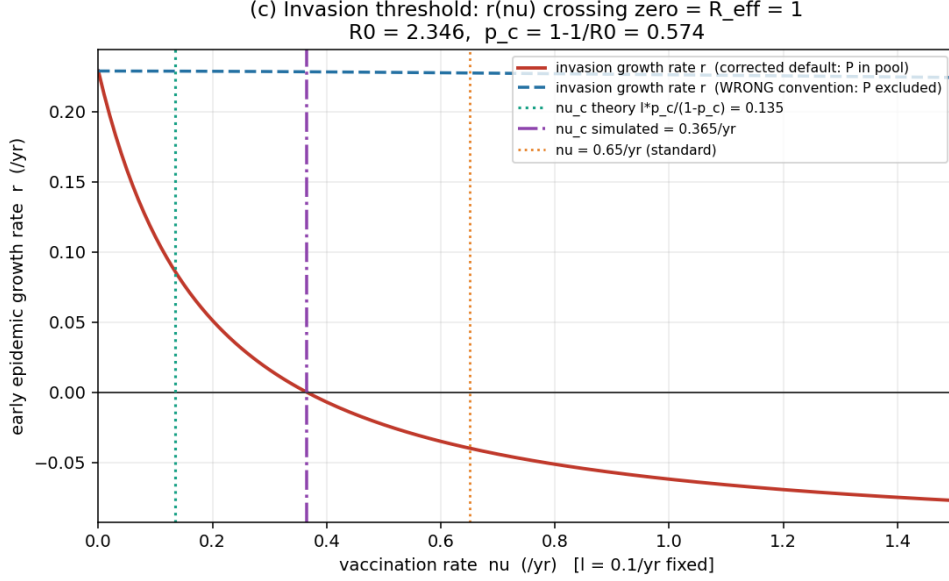


Figure 16: Q(c): invasion growth rate $r(\nu)$ crossing zero ($\mathcal{R}_{\text{eff}} = 1$) under the corrected convention (P in pool, finite $\nu_c \approx 0.366$); the wrong P -excluded convention (no crossing) is shown for contrast. Theory ν_c , simulated ν_c , and $\nu = 0.65$ are marked.

Vaccination carries an *explicit* \$348,618 over 80 yr and is capped by the waning ceiling 0.867, so it must re-vaccinate waned individuals indefinitely. Condoms are modelled as a permanent $0.5\times$ cut to $\mathcal{R}_0$ at no priced cost — but that cost is *unpriced*, not zero; a real behaviour-change programme has its own budget the project does not supply. A dollar-matched comparison would need a condom-programme cost we do not have. There is also the near-threshold caveat: condoms leave $\mathcal{R}_0$ only $\sim 17\%$ above one, and a higher baseline $\mathcal{R}_0$ (say a higher partner rate) would push condom- $\mathcal{R}_0$ clearly above one while raising vaccination's ν_c — both verdicts are $\mathcal{R}_0$ -sensitive near threshold, which is exactly why I report $\mathcal{R}_{\text{eff}}$ explicitly rather than just trajectories. The honest reading is a near-tie on epidemiological outcome, with vaccination marginally ahead for strictly crossing the threshold, and the two not directly comparable on cost as priced here.

2.9 Part 2 summary

On the sIC model, with γ recalibrated to a biologically sensible 0.1 yr^{-1} (giving $\mathcal{R}_0 \approx 2.35$) and the epidemic seeded in the infectious rather than the AIDS compartment, vaccination at $\nu = 0.65 \text{ yr}^{-1}$ controls the HIV epidemic: it drives $\mathcal{R}_{\text{eff}} \approx 0.31 < 1$, so the epidemic never ignites, and prevalence stays near zero instead of rising to the ~ 0.41 endemic plateau of the baseline. It is cost-effective ($\sim \$102$ per infection averted at 30 yr, within the published Garnett/Stover range), though the standard rate over-vaccinates somewhat relative to the cost-effective knee at $\nu \approx 0.18 \text{ yr}^{-1}$. Against condom promotion the two are roughly comparable on epidemiological outcome — both essentially control the epidemic, vaccination crossing the threshold strictly while condoms hover just above it — but they are not directly comparable on cost as priced here, and both verdicts are sensitive to the baseline $\mathcal{R}_0$ near threshold. All of these conclusions rest on the documented modelling assumptions, in particular the γ recalibration and the partner-pool convention that gives vaccination its herd-immunity effect.

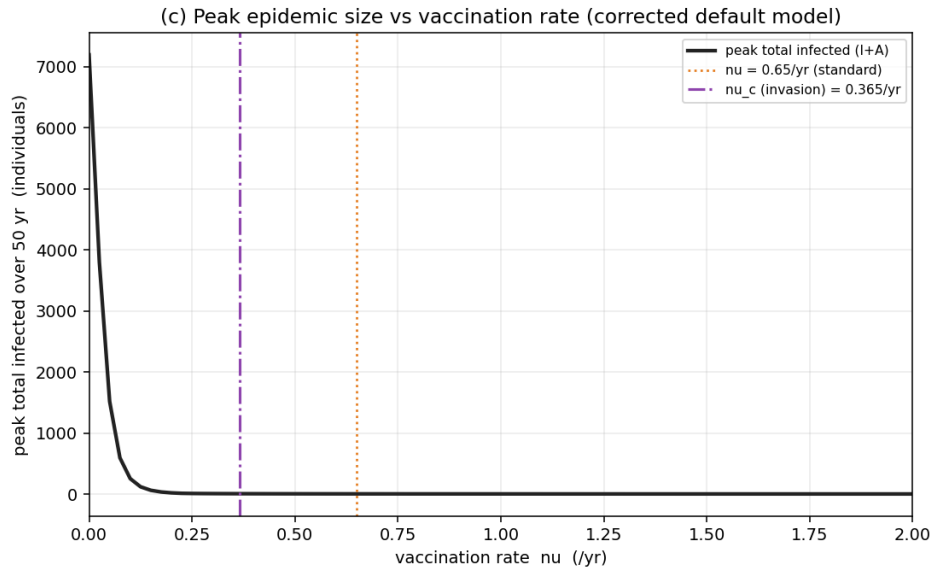


Figure 17: $Q(c)$: peak epidemic size versus ν . Above the threshold the epidemic is suppressed.

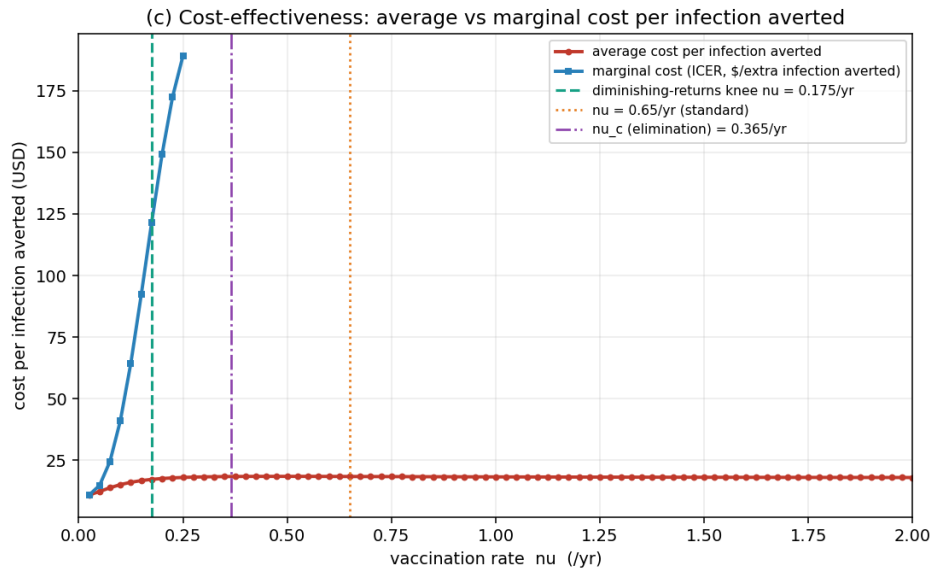


Figure 18: $Q(c)$: average and marginal (incremental) cost per infection averted versus ν , with the diminishing-returns knee at $\nu \approx 0.175$ marked.

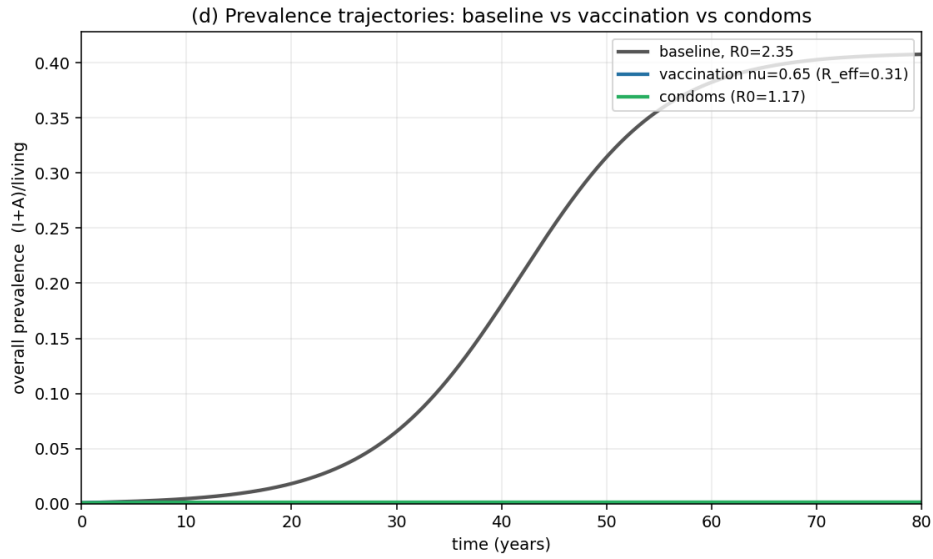


Figure 19: Q(d): overall prevalence trajectories for baseline, vaccination, and condoms. Both interventions collapse the epidemic; the baseline rises to a ~ 0.41 plateau.

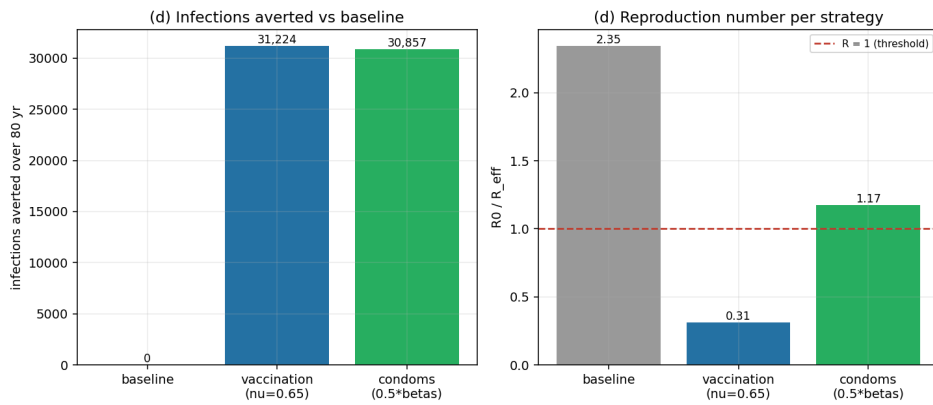


Figure 20: Q(d): infections averted and reproduction number per strategy. Vaccination crosses $R_{\text{eff}} < 1$ strictly; condoms sit just above $R_0 = 1$.

3 Reproducibility and code

All models and analyses are in Python (NumPy/SciPy/Matplotlib). Part 1 lives in `part1_glucose/` (`cobelli.py` core; `ex1_ivgitt.py... ex7_icecream.py`; `verify_core.py`); Part 2 in `part2_hiv/` (`sic.py` core; `q_a... q_d` analyses; `verify_sic.py`). Each core ships a verification harness that checks the basal steady state / conservation of individuals and agreement between the LSODA and Radau integrators. Octave (`.m`) cross-check scripts for the two cores are provided in `octave_crosscheck/`.

4 References

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